MILITARY SPECIFICATION SHEET

PLASTIC SHEETS, LAMINATED, THERMOSETTING ELECTRICAL-INSULATING SHEET,

POLYESTER GLASS-MAT GRADE GPO-N2 (CLASS 130)

The complete requirements for procuring the plastic sheets described herein shall consist of this document and the lastest issue of Specification MIL-P-24364(SHIPS).

The plastic laminated sheets shall be as follows:

Grade GPO-N2 Polyester Resin Glass-Mat, Low Flammability and Self-extinguishing properties.

Requirements: The property value requirements shall be as shown in tables I and II.

FSC 9330

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Table I - Property value requirements for

Qualification Test for Polyester Resin Glass-Mat Grade GPO-N2

Class 130

Properties to be tested	Test paragraph	Number of specimens for each thickness	Conditioning (see 4.6)	Units of value	Values required for each thickness: 1/16 inch	1/8 inch	1/2 inch
Thickness	4.7.3	1	As received	Inch	±0.008	±0.010	±0.024
Dielectric breakdown parallel to laminations	4.7.5	1 (short-time) 4 (step-by-step) 1 (short-time) 4 (step-by-step)	As received 336/50 water	Average KV-min. Average KV-min.	25 15	40 25	40 30
Dielectric constant 60 hertz 1 megahertz	4.7.6	4 4 4 4	As received 24/23 water As received 24/23 water	Maximum	---- ---- ---- ----	5.0 6.0 4.5 5.5	---- ---- ---- ----
Dissipation factor 60 hertz 1 megahertz	4.7.6	4 4 4 4	As received 24/23 water As received 24/23 water	Maximum	---- ---- ---- ----	0.05 0.15 0.03 0.06	---- ---- ---- ----
Arc resistance	4.7.7	4	As received 48/50 water	Seconds-min.	180 180	180 180	180 180
Wet tracking	4.7.8	4	As received	Minutes-min.	10 1/	10 1/	10
Impact strength, tested edgewise	4.7.9	4	48/50	Foot lbs per inch-min.	----	8	8
Flexural strength, tested flatwise (crosswise)	4.7.10	5	As received	P.S.I.-min.	16000	18000	18000
Bond strength	4.7.11	4 4	As received 48/50 water	Pounds-min.	---- ----	---- ----	850 800
Water absorption	4.7.12	4	1/105 + 24/23 water	Percent-max.	0.6	0.5	0.3
Flame resistance Ignition time Burning time	4.7.13	5	As received	Seconds-min. Seconds-max.	---- ----	---- ----	75 85
Thermal endurance	4.7.14			Temp. index °C. at 25,000 hrs. (min.)	----	130	----
Silicone content	4.7.15	1	168/185	ppm-max.	25	25	25
Machinability	4.7.16	1	As received	Pass or fail	----	----	Pass

1/ Specimens of the same material shall be built up to 0.250 inch.

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Table II - Property value requirements for

Quality conformance test polyester glass-mat for grade GPO-N2.

Properties to be tested	Test paragraph	Number of specimens for each thickness	Conditioning (see 4.6)	Units of value	Values required for each thickness range[1] (inches)				
					0.625 or less	0.0625-0.094	0.095-0.250	0.251-1.000	1.001-2.000
Warp or twist	4.7.4	1	As received	Percent-max.	2.50%	----	2.50%	0.50%	----
Dielectric breakdown parallel to to lamina-tions	4.7.5	1 (short-time) 4 (step-by-step)	48/50 water	Average KV-min.	15	15	25	30	35
Flexural strength tested, flat-wise, (cross-wise)	4.7.10	4	As received	P.S.I.-min.	16000	16000	18000	16000	14000
Bond strength	4.7.11	4	As received	Pounds-min.	----	----	----	850[2]	850
Flame resistance	4.7.13	5	As received						
Ignition time				Seconds-min.	----	----	----	75	----
Burning time				Seconds-max.	----	----	----	85	----

[1] These ranges applicable for the nominal thickness.

[2] For specimens 0.500 inch and above.

Preparing activity:
(Navy - SH)
(Project 9330-N446)

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